Sina Parsnejad

Raleigh, North Carolina 27612 | (517) 402-8575 | s.parsnejad@gmail.com

Work Authorization: U.S. Permanent Resident (**Green Card holder**)

Social: www.linkedin.com/in/sina-parsnejad | Portfolio: https://sparsnejad.github.io/Portfolio.pdf

PERSONAL STATEMENT

Ph.D.-trained **Analog IC Design Engineer** with expertise in **mixed signal circuits, power electronics, sensors, and brain-computer interfaces.** Industry experience at Texas Instruments includes the full IC lifecycle from specification and design to simulation, validation, and post-silicon testing on proprietary CMOS technologies. Proficient in **Cadence Design Suite, MATLAB, Verilog, and embedded systems,** with a strong record of translating advanced research into reliable, high-performance semiconductor solutions. Passionate about developing **next-generation solutions** that bridge the physical and digital worlds.

EXPERIENCE

Analog IC Design Engineer, *Texas Instruments,* 2022–2025

- Delivered 20 tape-outs for low-side and half-bridge driver IPs (UCC571xx, UCC276xx, UCC274xx, UCC278xx families) on 90nm BCD/CMOS, achieving ~60% smaller chip area on average vs. competitors, enabling ~30% lower product cost
- Designed analog building blocks (LDOs, bandgaps, opamps, ADCs/DACs, sensor interfaces) achieving 100% first-pass silicon success and enabling AEC-Q100 automotive qualification and design wins at Tier 1 OEMs
- Performed pre-silicon statistical and top-level sims, coordinated circuit layout implementation and post-layout sims, conducted automotive aging compliance testing, and in-laboratory post-silicon verification for low-side driver ICs
- Conducted quality assurance on TI projects, identifying and resolving design issues by performing post-silicon failure analysis and providing customer support for returned products
- Authored detailed design documentation, simulation reports, and design reviews for project milestones
- Performed outreach program as TI liaison for IEEE conferences and colleges

Research Assistant at *HAT lab,* *Michigan State University,* 2014–2022

- Led IRB-approved human trials with 40 participants for peripheral nerve stimulation device, demonstrating 85% recognition accuracy across 24 distinct tactile patterns and publishing findings in IEEE TBME, IEEE TbioCAS, and IEEE Haptics
- Led instrument design using EDA PCB design tools and collaborated with cross-functional team (chemical, software, electrical, mechanical) on grant proposals securing $2.5M in NIH funding
- Designed high-precision analog readout interface for nanopore biosensors on ON Semiconductor 0.6µm CMOS, achieving industry-leading 1pA noise floor for single-molecule detection

Freelance Artificial Intelligence Researcher, *Mercor,* 2025–Present

- Trained a generative AI model to tackle Olympiad-level engineering questions, enhancing problem-solving capabilities.
- Developed complex engineering questions, ensuring they met rigorous academic standards for evaluation.
- Graded and corrected answers to provide accurate feedback, fostering a deeper understanding of engineering concepts.

Venture Fellow, *Spartan Innovation Center,* 2018–2019

- Market research and business plan development for MSU startup company, MagPlasma, Inc
- Participated in pitch competitions and investor meetings to present and promote potential business solutions

Research Assistant at *Beta* Laboratory, *Boğaziçi University,* 2012–2014

- Designed analog ICs in UMC 180nm technology, including continuous-time Sigma-Delta ADCs

EDUCATION

Michigan State University, Ph.D. of Electrical Engineering, East Lansing, Michigan

- Supervisor: Professor Andrew J. Mason • July 2017–August 2022
- Thesis: Human augmentation using unobtrusive rapid real-time machine-to-human communication

Michigan State University, M.S. of Electrical Engineering, East Lansing, Michigan

- Supervisor: Professor Andrew J. Mason • September 2014–July 2017
- Thesis: Power-area-efficient rapid-response CMOS frontend for high-throughput ion channel sensor array microsystems

SKILLS

Proficient in IC design using Cadence Design Suite and Mentor Graphics from conception through statistical design, test, and post-silicon validation. Proficient in Python, MATLAB, C, and Verilog. Familiar with ARM embedded systems, NVIDIA Jetson, ML frameworks, FPGA (Altera MAX10), and PCB/PCBA design and testing. Experienced in Jira change management, root cause analysis (RCA), and failure mode analysis. Languages: English, Persian, Turkish, Azerbaijani (fluent); Japanese (intermediate).

ADDITIONAL EXPERIENCE

- Associate editor for IEEE Open Journal of Circuits and Systems (OJCAS), 2026-2027
- IEEE MWSCAS 2025 conference Chair of Tutorials
- Invited presentation at IEEE Colombian BioCas conference, 2020 and NCSU University ASSIST / IConS Speaker series, 2025

PUBLICATIONS

- "A review of electrotactile stimulation for machine-to-human communication" IEEE TBME, 2026
- "Distinguishable Multi-layer Frequency-modulated Waveforms for Peripheral Nerve Communication in Human Augmentation Applications" IEEE Haptics, 2026
- "Etacteme Machine-to-Human Communication A Linguistically-Inspired Experimentally-Derived Electrotactile Icon Generation Scheme" IEEE TBME, 2026